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Network and end-user equipment developers have been making steady progress in addressing these challenges over the last few years, but critical to the successful rollout is the development of test and measurement (T&M) equipment that enables developers to evaluate the performance and standards compliance of this new equipment in the lab, in production and in the field.

Therefore to get 5G equipment to market quickly, efficiently and economically requires T&M vendors to work closely with both the top-tier equipment makers and standards authorities to resolve their own set of technical challenges and provide a complete spectrum of T&M solutions that are both fit for purpose and have a low cost of ownership.

Right test solutions for 5G

Solutions for 5G testing challenges are both evolutionary and revolutionary. Since both 4G and 5G are based on orthogonal frequency-division multiplexing (OFDM), existing equipment designed for 4G can often be upgraded, in some cases just with software, to make it suitable for 5G. This is the case with signal generators and spectrum and signal analysers from leading, established T&M equipment vendors, who design and future-proof to reduce not only development costs for themselves, but notably also the total cost of ownership of their products for their customers.

In other cases, hardware evolution is required. This is the case, for example, with vector network analysers, which are now required to have true multiport test capabilities to support massive-MIMO. A top-of-the-range 5G analyser can now boast support for up to 24 ports in real time, which can be increased to 288 ports using a

switch matrix. This is important as massive-MIMO antenna arrays typically have 128 antenna elements or more. Simultaneous testing of LTE and 5G terminal devices, necessary for the network architecture favoured by the majority of network operators, also requires hardware enhancements, in part because over-the-air (OTA) measurement solutions need additional shielded chambers.

Where evolution of existing T&M equipment or designs is not feasible, however, technological revolution is required. A clear example in this category is a consequence of the introduction of new frequency bands. Implementation of components, chipsets, modules, wireless devices and base stations in the 28GHz or 39GHz band requires higher integration density and the use of active antenna systems to enable efficient beamforming.

Need for low path loss and small size lead to highly-integrated PCBs that include antennae, amplifiers and analogue phase shifters. As a result, radio frequency (RF) connectors are no longer available. This means that OTA test solutions have to be rolled out to replace existing conducted measurement methods.

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However, OTA testing at sub-6GHz frequencies is very difficult as far-field analysis would require very large chambers, measuring ten metres or more. One innovative solution to this problem is to make near-field measurements under far-field conditions. At cm/mmWave frequencies compact antenna test range (CATR) solutions apply, which create far-field conditions in close vicinity to the device under test.

CATR realises near-field to far-field transformation with a well-designed and highly-accurate produced mirror. Reflector weight significantly increases below 6GHz operation, since massive-MIMO antennae easily reach radiating dimensions greater than seventy or eighty centimetres. Cost, fabrication time and handling of large heavy mirrors become prohibitive. Using an antenna array, it is possible to create a planar field that allows for a quiet zone of one metre in diameter within a measurement distance of 1.5 metres, thus enabling much more convenient and cost-effective equipment to be used.

As mentioned, with the introduction of mMTC and URLLC in particular, 5G facilitates a whole new range of applications from automated driving to the Internet of Things (IoT) and cloud robotics. This broadens the field of T&M to encompass additional requirements such as safety and security, where reliability, deterministic low latency, authentication and encryption are critical. A T&M equipment supplier with products already in these markets and an experience in such topics as cybersecurity and radio monitoring will have much to contribute to 5G testing in these areas.

Moving forward

Clearly, 5G T&M is a complex and rapidly evolving topic. Successful T&M equipment suppliers will need to build on their success and experience in 4G, evolving the software and hardware, where feasible, to minimise costs both for themselves and their customers. They will need to leverage their close relationships with their customers and work together to develop solutions for 5G that will enable those customers to access the new markets first.

Taking an active role in driving the new standards from within the standards bodies, particularly 3rd Generation Partnership Project (3GPP), is providing the advanced insight that will also facilitate early market access.

Finally, as new 5G-enabled applications evolve, it is expected that today's single-source, broad-range T&M suppliers will have most to offer to companies that are keen to make early gains in developing markets. **EFY**